EXPERIMENT 2 — REPEATED STRATIFIED K-FOLD CROSS-VALIDATION RESULTS

As a complementary robustness analysis, we report a repeated stratified K-fold cross-validation protocol (7 folds $\times$ 5 repeats = 35 train/validation partitions, fixed subject-level 80/20 train/test split, split_random_state = 42) for both the HIS-RKHS and PPK, each combined with KNN and SVM classifiers, across the three HMM topologies evaluated (N3G3, N4G4, N5G5 = number of hidden states $\times$ number of GMM components per state). For each configuration, hyperparameters were first selected via an Optuna search using the repeated K-fold objective on the training partition only; we then report (i) the repeated stratified subject-level cross-validation performance (accuracy, balanced accuracy, AUC-ROC, and MCC, mean $\pm$ std across the 35 folds) under the selected hyperparameters, and (ii) the resulting held-out test-set evaluation, including balanced accuracy, sensitivity, specificity, AUC-ROC, MCC, and the confusion matrix. Unlike Experiment 1, this protocol does not perform a nested nor a bootstrap nor a label-permutation procedure; consequently, no bootstrap confidence interval or permutation p-value is available for these results, and dispersion is instead reported as mean $\pm$ standard deviation and the fold-level min/max.

Dataset N3G3

(HIS-RKHS)

KNN

Table 1: Repeated stratified subject-level evaluation (7 folds $\times$ 5 repeats = 35 train/test partitions), mean $\pm$ std over folds — (HIS-RKHS) – KNN, dataset N3G3. Selected hyperparameters (best inner-CV balanced accuracy 0.8214): sigma_kernel: 0.23381611146310066, k_neighbors: 1.

Metric	Mean ± Std	Min	Max
Accuracy	0.8207 ± 0.0914	0.6154	1.0000
Balanced Accuracy	0.8214 ± 0.0912	0.6310	1.0000
AUC-ROC	0.8214 ± 0.0912	0.6310	1.0000
MCC	0.6589 ± 0.1775	0.2829	1.0000

Table 2: Held-out test set evaluation (single subject-level 80/20 split, $n_{\text{test}} = 25$) — (HIS-RKHS) – KNN, dataset N3G3.

Metric	Value
Accuracy	0.9200
Balanced Accuracy	0.9199
Sensitivity	0.9231
Specificity	0.9167
AUC-ROC	0.9199
MCC	0.8397

Table 3: Test-set confusion matrix — (HIS-RKHS) – KNN, dataset N3G3.

	Pred. Control	Pred. ADHD
True Control	11	1
True ADHD	1	12

SVM

Table 4: Repeated stratified subject-level evaluation (7 folds × 5 repeats = 35 train/test partitions), mean ± std over folds — (HIS-RKHS) – SVM, dataset N3G3. Selected hyperparameters (best inner-CV balanced accuracy 0.8122): sigma_kernel=0.0476726, gamma=0.00325884, C=757.551.

Metric	Mean ± Std	Min	Max
Accuracy	0.8108 ± 0.0998	0.5385	1.0000
Balanced Accuracy	0.8122 ± 0.0969	0.5714	1.0000
AUC-ROC	0.9016 ± 0.0656	0.7381	1.0000
MCC	0.6539 ± 0.1792	0.2673	1.0000

Table 5: Held-out test set evaluation (single subject-level 80/20 split, $n_{\text{test}} = 25$) — (HIS-RKHS) – SVM, dataset N3G3.

Metric	Value
Accuracy	0.8400
Balanced Accuracy	0.8429
Sensitivity	0.7692
Specificity	0.9167
AUC-ROC	0.8942
MCC	0.6903

Table 6: Test-set confusion matrix — (HIS-RKHS) – SVM, dataset N3G3.

	Pred. Control	Pred. ADHD
True Control	11	1
True ADHD	3	10

Probability Product Kernel (PPK)

KNN

Table 7: Repeated stratified subject-level evaluation (7 folds $\times$ 5 repeats = 35 train/test partitions), mean $\pm$ std over folds — Probability Product Kernel (PPK) – KNN, dataset N3G3. Selected hyperparameters (best inner-CV balanced accuracy 0.8622): T=3, rho=0.708216, k_neighbors=1.

Metric	Mean ± Std	Min	Max
Accuracy	0.8614 ± 0.0999	0.6154	1.0000
Balanced Accuracy	0.8622 ± 0.0985	0.6310	1.0000
AUC-ROC	0.8622 ± 0.0985	0.6310	1.0000
MCC	0.7393 ± 0.1909	0.2829	1.0000

Table 8: Held-out test set evaluation (single subject-level 80/20 split, $n_{\text{test}} = 25$) — Probability Product Kernel (PPK) – KNN, dataset N3G3.

Metric	Value
Accuracy	0.7600
Balanced Accuracy	0.7596
Sensitivity	0.7692
Specificity	0.7500
AUC-ROC	0.7596
MCC	0.5192

Table 9: Test-set confusion matrix — Probability Product Kernel (PPK) – KNN, dataset N3G3.

	Pred. Control	Pred. ADHD
True Control	9	3
True ADHD	3	10

SVM

Table 10: Repeated stratified subject-level evaluation (7 folds $\times$ 5 repeats = 35 train/test partitions), mean $\pm$ std over folds — Probability Product Kernel (PPK) – SVM, dataset N3G3. Selected hyperparameters (best inner-CV balanced accuracy 0.9010): T=6, rho=0.860921, gamma=0.0001577, C=15.8699.

Metric	Mean ± Std	Min	Max
Accuracy	0.9000 ± 0.0732	0.7143	1.0000
Balanced Accuracy	0.9010 ± 0.0727	0.7143	1.0000
AUC-ROC	0.9674 ± 0.0465	0.8333	1.0000
MCC	0.8112 ± 0.1421	0.4286	1.0000

Table 11: Held-out test set evaluation (single subject-level 80/20 split, $n_{\text{test}} = 25$) — Probability Product Kernel (PPK) – SVM, dataset N3G3.

Metric	Value
Accuracy	0.8400
Balanced Accuracy	0.8429
Sensitivity	0.7692
Specificity	0.9167
AUC-ROC	0.9487
MCC	0.6903

Table 12: Test-set confusion matrix — Probability Product Kernel (PPK) – SVM, dataset N3G3.

	Pred. Control	Pred. ADHD
True Control	11	1
True ADHD	3	10

Dataset N4G4

(HIS-RKHS)

KNN

Table 13: Repeated stratified subject-level evaluation (7 folds × 5 repeats = 35 train/test partitions), mean ± std over folds — (HIS-RKHS) – KNN, dataset N4G4. Selected hyperparameters (best inner-CV balanced accuracy 0.6197): sigma_kernel: 0.021058967196437517, k_neighbors: 5.

Metric	Mean ± Std	Min	Max
Accuracy	0.6181 ± 0.1232	0.4286	0.8462
Balanced Accuracy	0.6197 ± 0.1231	0.4286	0.8452
AUC-ROC	0.6253 ± 0.1846	0.2857	0.9762
MCC	0.2573 ± 0.2582	-0.1741	0.6905

Table 14: Held-out test set evaluation (single subject-level 80/20 split, $n_{\text{test}} = 25$) — (HIS-RKHS) – KNN, dataset N4G4.

Metric	Value
Accuracy	0.5600
Balanced Accuracy	0.5481
Sensitivity	0.8462
Specificity	0.2500
AUC-ROC	0.6250
MCC	0.1201

Table 15: Test-set confusion matrix — (HIS-RKHS) – KNN, dataset N4G4.

	Pred. Control	Pred. ADHD
True Control	3	9
True ADHD	2	11

SVM

Table 16: Repeated stratified subject-level evaluation (7 folds × 5 repeats = 35 train/test partitions), mean ± std over folds —(HIS-RKHS) – SVM, dataset N4G4. Selected hyperparameters (best inner-CV balanced accuracy 0.6520): sigma_kernel=0.012351, gamma=0.00516308, C=14.807.

Metric	Mean ± Std	Min	Max
Accuracy	0.6507 ± 0.1168	0.4286	0.8571
Balanced Accuracy	0.6520 ± 0.1178	0.4286	0.8571
AUC-ROC	0.7168 ± 0.1382	0.4490	0.9524
MCC	0.3260 ± 0.2400	-0.1491	0.7319

Table 17: Held-out test set evaluation (single subject-level 80/20 split, $n_{\text{test}} = 25$) — (HIS-RKHS) – SVM, dataset N4G4.

Metric	Value
Accuracy	0.6000
Balanced Accuracy	0.5962
Sensitivity	0.6923
Specificity	0.5000
AUC-ROC	0.6282
MCC	0.1961

Table 18: Test-set confusion matrix — (HIS-RKHS) – SVM, dataset N4G4.

	Pred. Control	Pred. ADHD
True Control	6	6
True ADHD	4	9

Probability Product Kernel (PPK)

KNN

Table 19: Repeated stratified subject-level evaluation (7 folds × 5 repeats = 35 train/test partitions), mean ± std over folds — Probability Product Kernel (PPK) – KNN, dataset N4G4. Selected hyperparameters (best inner-CV balanced accuracy 0.7493): T=12, rho=0.179386, k_neighbors=1.

Metric	Mean ± Std	Min	Max
Accuracy	0.7504 ± 0.1112	0.5000	0.9286
Balanced Accuracy	0.7493 ± 0.1111	0.5000	0.9286
AUC-ROC	0.7493 ± 0.1111	0.5000	0.9286
MCC	0.5200 ± 0.2261	0.0000	0.8660

Table 20: Held-out test set evaluation (single subject-level 80/20 split, $n_{\text{test}} = 25$) — Probability Product Kernel (PPK) – KNN, dataset N4G4.

Metric	Value
Accuracy	0.7200
Balanced Accuracy	0.7212
Sensitivity	0.6923
Specificity	0.7500
AUC-ROC	0.7212
MCC	0.4423

Table 21: Test-set confusion matrix — Probability Product Kernel (PPK) – KNN, dataset N4G4.

	Pred. Control	Pred. ADHD
True Control	9	3
True ADHD	4	9

SVM

Table 22: Repeated stratified subject-level evaluation (7 folds × 5 repeats = 35 train/test partitions), mean ± std over folds — Probability Product Kernel (PPK) – SVM, dataset N4G4. Selected hyperparameters (best inner-CV balanced accuracy 0.7636): T=1, rho=0.100585, gamma=0.000374306, C=6.44559.

Metric	Mean ± Std	Min	Max
Accuracy	0.7630 ± 0.1161	0.4615	1.0000
Balanced Accuracy	0.7636 ± 0.1142	0.4881	1.0000
AUC-ROC	0.8196 ± 0.1022	0.5714	1.0000
MCC	0.5407 ± 0.2288	-0.0329	1.0000

Table 23: Held-out test set evaluation (single subject-level 80/20 split, $n_{\text{test}} = 25$) — Probability Product Kernel (PPK) – SVM, dataset N4G4.

Metric	Value
Accuracy	0.7600
Balanced Accuracy	0.7596
Sensitivity	0.7692
Specificity	0.7500
AUC-ROC	0.8205
MCC	0.5192

Table 24: Test-set confusion matrix — Probability Product Kernel (PPK) – SVM, dataset N4G4.

	Pred. Control	Pred. ADHD
True Control	9	3
True ADHD	3	10

Dataset N5G5

(HIS-RKHS)

KNN

Table 25: Repeated stratified subject-level evaluation (7 folds × 5 repeats = 35 train/test partitions), mean ± std over folds — (HIS-RKHS) – KNN, dataset N5G5. Selected hyperparameters (best inner-CV balanced accuracy 0.5942): sigma_kernel: 0.0116735651057811, k_neighbors: 1.

Metric	Mean ± Std	Min	Max
Accuracy	0.5951 ± 0.1458	0.2308	0.9286
Balanced Accuracy	0.5942 ± 0.1463	0.2381	0.9286
AUC-ROC	0.5942 ± 0.1463	0.2381	0.9286
MCC	0.1890 ± 0.3048	-0.5367	0.8660

Table 26: Held-out test set evaluation (single subject-level 80/20 split, $n_{\text{test}} = 25$) —(HIS-RKHS) – KNN, dataset N5G5.

Metric	Value
Accuracy	0.6800
Balanced Accuracy	0.6763
Sensitivity	0.7692
Specificity	0.5833
AUC-ROC	0.6763
MCC	0.3595

Table 27: Test-set confusion matrix — (HIS-RKHS) – KNN, dataset N5G5.

	Pred. Control	Pred. ADHD
True Control	7	5
True ADHD	3	10

SVM

Table 28: Repeated stratified subject-level evaluation (7 folds × 5 repeats = 35 train/test partitions), mean ± std over folds —(HIS-RKHS) – SVM, dataset N5G5. Selected hyperparameters (best inner-CV balanced accuracy 0.6071): sigma_kernel=0.0139534, gamma=0.00904554, C=16.4522.

Metric	Mean ± Std	Min	Max
Accuracy	0.6071 ± 0.1273	0.2857	0.9231
Balanced Accuracy	0.6071 ± 0.1274	0.2857	0.9167
AUC-ROC	0.6662 ± 0.1590	0.2653	0.9524
MCC	0.2397 ± 0.2697	-0.4472	0.8539

Table 29: Held-out test set evaluation (single subject-level 80/20 split, $n_{\text{test}} = 25$) — (HIS-RKHS) – SVM, dataset N5G5.

Metric	Value
Accuracy	0.7600
Balanced Accuracy	0.7660
Sensitivity	0.6154
Specificity	0.9167
AUC-ROC	0.8590
MCC	0.5538

Table 30: Test-set confusion matrix — (HIS-RKHS) – SVM, dataset N5G5.

	Pred. Control	Pred. ADHD
True Control	11	1
True ADHD	5	8

Probability Product Kernel (PPK)

KNN

Table 31: Repeated stratified subject-level evaluation (7 folds × 5 repeats = 35 train/test partitions), mean ± std over folds — Probability Product Kernel (PPK) – KNN, dataset N5G5. Selected hyperparameters (best inner-CV balanced accuracy 0.6949): T=6, rho=0.419953, k_neighbors=3.

Metric	Mean ± Std	Min	Max
Accuracy	0.6945 ± 0.1166	0.4286	0.9231
Balanced Accuracy	0.6949 ± 0.1178	0.4286	0.9167
AUC-ROC	0.7390 ± 0.1425	0.3690	0.9592
MCC	0.4070 ± 0.2439	-0.1429	0.8539

Table 32: Held-out test set evaluation (single subject-level 80/20 split, $n_{\text{test}} = 25$) — Probability Product Kernel (PPK) – KNN, dataset N5G5.

Metric	Value
Accuracy	0.8800
Balanced Accuracy	0.8814
Sensitivity	0.8462
Specificity	0.9167
AUC-ROC	0.9359
MCC	0.7628

Table 33: Test-set confusion matrix — Probability Product Kernel (PPK) – KNN, dataset N5G5.

	Pred. Control	Pred. ADHD
True Control	11	1
True ADHD	2	11

SVM

Table 34: Repeated stratified subject-level evaluation (7 folds × 5 repeats = 35 train/test partitions), mean ± std over folds — Probability Product Kernel (PPK) – SVM, dataset N5G5. Selected hyperparameters (best inner-CV balanced accuracy 0.7490): T=11, rho=0.23619, gamma=0.000402158, C=2.77846.

Metric	Mean ± Std	Min	Max
Accuracy	0.7480 ± 0.0981	0.5000	1.0000
Balanced Accuracy	0.7490 ± 0.0976	0.5000	1.0000
AUC-ROC	0.7959 ± 0.1144	0.4898	1.0000
MCC	0.5151 ± 0.1992	0.0000	1.0000

Table 35: Held-out test set evaluation (single subject-level 80/20 split, $n_{\text{test}} = 25$) — Probability Product Kernel (PPK) – SVM, dataset N5G5.

Metric	Value
Accuracy	0.7200
Balanced Accuracy	0.7179
Sensitivity	0.7692
Specificity	0.6667
AUC-ROC	0.7788
MCC	0.4387

Table 36: Test-set confusion matrix — Probability Product Kernel (PPK) – SVM, dataset N5G5.

	Pred. Control	Pred. ADHD
True Control	8	4
True ADHD	3	10

Table 37: Experiment 2 (repeated stratified K-fold CV) — comparison across HMM topologies N3F7, N4F7, and N5F7 (number of hidden states $\times$ number of GMM components per state). CV columns report mean $\pm$ std over 35 folds (7 folds $\times$ 5 repeats); Test columns report the held-out subject-level test-set evaluation (single 80/20 split). Sensitivity and specificity are computed from the test-set confusion matrix.

		Repeated CV (35 folds)			Held-out Test Set				
Configuration	Method	Bal. Acc.	AUC	MCC	Acc.	Bal. Acc.	Sens.	Spec.	MCC
N3G3									
	HIS-KNN	0.8214 ± 0.0912	0.8214 ± 0.0912	0.6589 ± 0.1775	0.9200	0.9199	0.9231	0.9167	0.8397
	HIS-SVM	0.8122 ± 0.0969	0.9016 ± 0.0656	0.6539 ± 0.1792	0.8400	0.8429	0.7692	0.9167	0.6903
	PPK-KNN	0.8622 ± 0.0985	0.8622 ± 0.0985	0.7393 ± 0.1909	0.7600	0.7596	0.7692	0.7500	0.5192
	PPK-SVM	0.9010 ± 0.0727	0.9674 ± 0.0465	0.8112 ± 0.1421	0.8400	0.8429	0.7692	0.9167	0.6903
N4G4									
	HIS-KNN	0.6197 ± 0.1231	0.6253 ± 0.1846	0.2573 ± 0.2582	0.5600	0.5481	0.8462	0.2500	0.1201
	HIS-SVM	0.6520 ± 0.1178	0.7168 ± 0.1382	0.3260 ± 0.2400	0.6000	0.5962	0.6923	0.5000	0.1961
	PPK-KNN	0.7493 ± 0.1111	0.7493 ± 0.1111	0.5200 ± 0.2261	0.7200	0.7212	0.6923	0.7500	0.4423
	PPK-SVM	0.7636 ± 0.1142	0.8196 ± 0.1022	0.5407 ± 0.2288	0.7600	0.7596	0.7692	0.7500	0.5192
N5G5									
	HIS-KNN	0.5942 ± 0.1463	0.5942 ± 0.1463	0.1890 ± 0.3048	0.6800	0.6763	0.7692	0.5833	0.3595
	HIS-SVM	0.6071 ± 0.1274	0.6662 ± 0.1590	0.2397 ± 0.2697	0.7600	0.7660	0.6154	0.9167	0.5538
	PPK-KNN	0.6949 ± 0.1178	0.7390 ± 0.1425	0.4070 ± 0.2439	0.8800	0.8814	0.8462	0.9167	0.7628
	PPK-SVM	0.7490 ± 0.0976	0.7959 ± 0.1144	0.5151 ± 0.1992	0.7200	0.7179	0.7692	0.6667	0.4387